

Etapa 1. El contexto

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Exploración de la base de datos

```
library(tidyverse)
```

```
— Attaching core tidyverse packages — tidyverse 2.0.0
```

```
✓ dplyr      1.1.4    ✓ readr      2.1.5
✓ forcats    1.0.1    ✓ stringr    1.5.2
✓ ggplot2    4.0.0    ✓ tibble     3.3.0
✓ lubridate  1.9.4    ✓ tidyr      1.3.1
✓ purrr      1.1.0
```

```
— Conflicts — tidyverse_conflicts()
```

```
✗ dplyr::filter() masks stats::filter()
```

```
✗ dplyr::lag() masks stats::lag()
```

```
ℹ Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
library(here)
```

```
here() starts at /Users/emmahuerta/Documents/GitHub/MA1001B
```

```
M19 <- read_csv(here("data", "Datos_molec_2019.csv"))
```

```
Warning: One or more parsing issues, call `problems()` on your data frame for details,
```

```
e.g.:
```

```
dat <- vroom(...)
problems(dat)
```

```
Rows: 1980 Columns: 108
```

```
— Column specification
```

```
Delimiter: ","
```

```
chr (15): cd, periodo, folio, entidad, n_ren_ele, p5_6esp, p9_5esp,
p12_9esp...
```

```
dbl (91): control, viv_sel, num_hog, hog_mud, p1, p2, p3_1, p3_2, p3_3,
p3_4...
```

```
lgl (2): p6_6esp, p11_6esp
```

```
ℹ Use `spec()` to retrieve the full column specification for this data.
```

```
ℹ Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```
M20 <- read_csv(here("data", "Datos_molec_2020.csv"))
```

Warning: One or more parsing issues, call `problems()` on your data frame for details,

e.g.:

```
dat <- vroom(...)
problems(dat)
```

Rows: 2010 Columns: 108

— Column specification

Delimiter: ","

chr (11): folio, p5_6esp, p6_6esp, p11_6esp, p12_9esp, p15_5esp, p17_6esp, p...

dbl (96): entidad, control, viv_sel, num_hog, hog_mud, n_ren_el, cd, periodo...

lgl (1): p9_5esp

i Use `spec()` to retrieve the full column specification for this data.

i Specify the column types or set `show\_col\_types = FALSE` to quiet this message.

```
str(M19)
```

```
spc_tbl_ [1,980 × 108] (S3: spec_tbl_df/tbl_df/tbl/data.frame)
```

```
$ cd      : chr [1:1980] "14" "14" "14" "14" ...
$ periodo : chr [1:1980] "0219" "0219" "0219" "0219" ...
$ folio    : chr [1:1980] "11B167" "11B167" "11B167" "11B167" ...
$ entidad  : chr [1:1980] "01" "01" "01" "01" ...
$ control  : num [1:1980] 40006 40006 40006 40006 40131 ...
$ viv_sel  : num [1:1980] 3 2 1 4 3 4 2 1 1 2 ...
$ num_hog  : num [1:1980] 1 1 1 1 1 1 1 1 1 1 ...
$ hog_mud  : num [1:1980] 0 0 0 0 0 0 0 0 0 0 ...
$ n_ren_ele: chr [1:1980] "02" "01" "01" "02" ...
$ p1       : num [1:1980] 1 1 1 1 1 1 1 1 1 1 ...
$ p2       : num [1:1980] 2 2 2 2 2 1 1 1 2 1 ...
$ p3_1     : num [1:1980] 2 2 2 2 2 2 1 2 2 2 ...
$ p3_2     : num [1:1980] 2 2 2 2 2 1 2 2 2 2 ...
$ p3_3     : num [1:1980] 2 2 1 2 2 1 2 1 2 1 ...
$ p3_4     : num [1:1980] 2 2 2 2 2 2 2 2 2 2 ...
$ p3_5     : num [1:1980] 2 2 2 2 2 1 1 2 2 2 ...
$ p4       : num [1:1980] 0 0 0 0 0 0 1 0 0 0 ...
$ p5       : num [1:1980] 0 0 0 0 0 0 4 0 0 0 ...
$ p5_6esp  : chr [1:1980] NA NA NA NA ...
$ p6_1     : num [1:1980] 0 0 0 0 0 0 2 0 0 0 ...
$ p6_2     : num [1:1980] 0 0 0 0 0 0 2 0 0 0 ...
$ p6_3     : num [1:1980] 0 0 0 0 0 0 1 0 0 0 ...
$ p6_4     : num [1:1980] 0 0 0 0 0 0 2 0 0 0 ...
$ p6_5     : num [1:1980] 0 0 0 0 0 0 2 0 0 0 ...
$ p6_6     : num [1:1980] 0 0 0 0 0 0 2 0 0 0 ...
$ p6_6esp  : logi [1:1980] NA NA NA NA NA NA ...
```

```

$ p7      : num [1:1980] 0 0 0 0 0 0 2 0 0 0 ...
$ p7_3    : num [1:1980] 0 0 0 0 0 0 0 0 0 0 ...
$ p8_1    : num [1:1980] 0 0 0 0 0 0 2 0 0 0 ...
$ p8_2    : num [1:1980] 0 0 0 0 0 0 1 0 0 0 ...
$ p9      : num [1:1980] 0 0 0 0 0 0 1 0 0 0 ...
$ p9_5esp : chr [1:1980] NA NA NA NA ...
$ p10     : num [1:1980] 0 0 0 0 0 2 0 0 0 0 ...
$ p11     : num [1:1980] 0 0 0 0 0 4 0 0 0 0 ...
$ p11_6esp : logi [1:1980] NA NA NA NA NA NA ...
$ p12_1   : num [1:1980] 0 0 0 0 0 2 0 0 0 0 ...
$ p12_2   : num [1:1980] 0 0 0 0 0 1 0 0 0 0 ...
$ p12_3   : num [1:1980] 0 0 0 0 0 2 0 0 0 0 ...
$ p12_4   : num [1:1980] 0 0 0 0 0 2 0 0 0 0 ...
$ p12_5   : num [1:1980] 0 0 0 0 0 2 0 0 0 0 ...
$ p12_6   : num [1:1980] 0 0 0 0 0 1 0 0 0 0 ...
$ p12_7   : num [1:1980] 0 0 0 0 0 2 0 0 0 0 ...
$ p12_8   : num [1:1980] 0 0 0 0 0 2 0 0 0 0 ...
$ p12_9   : num [1:1980] 0 0 0 0 0 2 0 0 0 0 ...
$ p12_9esp : chr [1:1980] NA NA NA NA ...
$ p13     : num [1:1980] 0 0 0 0 0 2 0 0 0 0 ...
$ p13_3   : num [1:1980] 0 0 0 0 0 0 0 0 0 0 ...
$ p14_1   : num [1:1980] 0 0 0 0 0 2 0 0 0 0 ...
$ p14_2   : num [1:1980] 0 0 0 0 0 1 0 0 0 0 ...
$ p15     : num [1:1980] 0 0 0 0 0 1 0 0 0 0 ...
$ p15_5esp : chr [1:1980] NA NA NA NA ...
$ p16     : num [1:1980] 0 0 1 0 0 1 0 3 0 2 ...
$ p17     : num [1:1980] 0 0 4 0 0 4 0 3 0 3 ...
$ p17_6esp : chr [1:1980] NA NA NA NA ...
$ p18_1   : num [1:1980] 0 0 1 0 0 1 0 2 0 2 ...
$ p18_2   : num [1:1980] 0 0 1 0 0 1 0 1 0 1 ...
$ p18_3   : num [1:1980] 0 0 1 0 0 2 0 2 0 2 ...
$ p18_4   : num [1:1980] 0 0 1 0 0 2 0 2 0 2 ...
$ p18_5   : num [1:1980] 0 0 1 0 0 1 0 1 0 1 ...
$ p19     : num [1:1980] 0 0 2 0 0 2 0 2 0 2 ...
$ p19_3   : num [1:1980] 0 0 0 0 0 0 0 0 0 0 ...
$ p20_1   : num [1:1980] 0 0 1 0 0 2 0 2 0 2 ...
$ p20_2   : num [1:1980] 0 0 2 0 0 1 0 1 0 1 ...
$ p21     : num [1:1980] 0 0 1 0 0 2 0 2 0 2 ...
$ p21_5esp : chr [1:1980] NA NA NA NA ...
$ p22     : num [1:1980] 0 0 0 0 0 0 0 0 0 0 ...
$ p23_1   : num [1:1980] 0 0 0 0 0 0 0 0 0 0 ...
$ p23_2   : num [1:1980] 0 0 0 0 0 0 0 0 0 0 ...
$ p24     : num [1:1980] 0 0 0 0 0 5 2 0 0 0 ...
$ p25     : num [1:1980] 0 0 0 0 0 3 3 0 0 0 ...
$ p25_6esp : chr [1:1980] NA NA NA NA ...
$ p26     : num [1:1980] 0 0 15 0 0 10 60 15 0 20 ...
$ p27     : num [1:1980] 0 0 2 0 0 2 2 2 0 2 ...
$ p28     : num [1:1980] 0 0 0 0 0 0 0 0 0 0 ...
$ p28_7esp : chr [1:1980] NA NA NA NA ...
$ p29     : num [1:1980] 0 0 2 0 0 4 3 2 0 3 ...

```

```

$ p30      : num [1:1980] 0 0 2 0 0 4 3 3 0 3 ...
$ p31      : num [1:1980] 0 0 2 0 0 2 1 2 0 2 ...
$ p32      : num [1:1980] 3 5 0 5 2 0 0 0 1 0 ...
$ p32_6esp : chr [1:1980] NA NA NA NA ...
$ p33_1     : num [1:1980] 2 2 2 2 2 2 2 2 2 2 ...
$ p33_2     : num [1:1980] 2 2 2 2 2 2 2 2 2 2 ...
$ p33_3     : num [1:1980] 2 2 2 2 2 2 2 2 2 2 ...
$ p33_4     : num [1:1980] 2 2 2 2 2 2 2 2 2 2 ...
$ p34_1     : num [1:1980] 2 2 2 2 3 1 1 3 3 3 ...
$ p34_2     : num [1:1980] 2 1 1 1 3 1 1 3 3 1 ...
$ p34_3     : num [1:1980] 2 2 2 1 3 1 1 3 3 1 ...
$ p34_3_1   : num [1:1980] 0 0 0 1 0 2 2 0 0 1 ...
$ p34_4     : num [1:1980] 2 1 1 1 1 1 1 1 3 1 ...
$ p34_4_1   : num [1:1980] 0 1 1 1 1 1 1 1 0 1 ...
$ p35       : num [1:1980] 1 1 1 1 1 1 1 1 1 1 ...
$ p36_1     : num [1:1980] 2 1 1 1 3 1 1 3 3 1 ...
$ p36_2     : num [1:1980] 2 1 2 1 3 1 1 3 3 1 ...
$ p36_3     : num [1:1980] 2 1 2 1 3 1 1 3 3 1 ...
$ p36_4     : num [1:1980] 2 2 2 2 3 1 1 3 3 1 ...
$ factor    : num [1:1980] 18795 15188 22781 14041 16030 ...
$ h_lec     : num [1:1980] 4 4 3 4 4 1 1 1 4 1 ...
$ mat_lec   : num [1:1980] 4 4 3 4 4 3 2 3 4 3 ...
$ perslec   : num [1:1980] 2 2 1 2 2 1 1 1 2 1 ...

```

[list output truncated]

```

- attr(*, "spec")=
.. cols(
..   cd = col_character(),
..   periodo = col_character(),
..   folio = col_character(),
..   entidad = col_character(),
..   control = col_double(),
..   viv_sel = col_double(),
..   num_hog = col_double(),
..   hog_mud = col_double(),
..   n_ren_ele = col_character(),
..   p1 = col_double(),
..   p2 = col_double(),
..   p3_1 = col_double(),
..   p3_2 = col_double(),
..   p3_3 = col_double(),
..   p3_4 = col_double(),
..   p3_5 = col_double(),
..   p4 = col_double(),
..   p5 = col_double(),
..   p5_6esp = col_character(),
..   p6_1 = col_double(),
..   p6_2 = col_double(),
..   p6_3 = col_double(),
..   p6_4 = col_double(),
..   p6_5 = col_double(),

```

```
.. p6_6 = col_double(),
.. p6_6esp = col_logical(),
.. p7 = col_double(),
.. p7_3 = col_double(),
.. p8_1 = col_double(),
.. p8_2 = col_double(),
.. p9 = col_double(),
.. p9_5esp = col_character(),
.. p10 = col_double(),
.. p11 = col_double(),
.. p11_6esp = col_logical(),
.. p12_1 = col_double(),
.. p12_2 = col_double(),
.. p12_3 = col_double(),
.. p12_4 = col_double(),
.. p12_5 = col_double(),
.. p12_6 = col_double(),
.. p12_7 = col_double(),
.. p12_8 = col_double(),
.. p12_9 = col_double(),
.. p12_9esp = col_character(),
.. p13 = col_double(),
.. p13_3 = col_double(),
.. p14_1 = col_double(),
.. p14_2 = col_double(),
.. p15 = col_double(),
.. p15_5esp = col_character(),
.. p16 = col_double(),
.. p17 = col_double(),
.. p17_6esp = col_character(),
.. p18_1 = col_double(),
.. p18_2 = col_double(),
.. p18_3 = col_double(),
.. p18_4 = col_double(),
.. p18_5 = col_double(),
.. p19 = col_double(),
.. p19_3 = col_double(),
.. p20_1 = col_double(),
.. p20_2 = col_double(),
.. p21 = col_double(),
.. p21_5esp = col_character(),
.. p22 = col_double(),
.. p23_1 = col_double(),
.. p23_2 = col_double(),
.. p24 = col_double(),
.. p25 = col_double(),
.. p25_6esp = col_character(),
.. p26 = col_double(),
.. p27 = col_double(),
.. p28 = col_double(),
```

```

.. p28_7esp = col_character(),
.. p29 = col_double(),
.. p30 = col_double(),
.. p31 = col_double(),
.. p32 = col_double(),
.. p32_6esp = col_character(),
.. p33_1 = col_double(),
.. p33_2 = col_double(),
.. p33_3 = col_double(),
.. p33_4 = col_double(),
.. p34_1 = col_double(),
.. p34_2 = col_double(),
.. p34_3 = col_double(),
.. p34_3_1 = col_double(),
.. p34_4 = col_double(),
.. p34_4_1 = col_double(),
.. p35 = col_double(),
.. p36_1 = col_double(),
.. p36_2 = col_double(),
.. p36_3 = col_double(),
.. p36_4 = col_double(),
.. factor = col_double(),
.. h_lec = col_double(),
.. mat_lec = col_double(),
.. perslec = col_double(),
.. l_formato = col_double(),
.. r_formato = col_double(),
.. p_formato = col_double(),
.. persleclrp = col_double(),
.. edad = col_double(),
.. sexo = col_double(),
.. anio = col_double(),
.. nivel = col_double(),
.. cond_activ = col_character()
.. )
- attr(*, "problems")=<externalptr>

```

1980 observaciones con 108 variables. La mayoría de las variables son cualitativas, pero están insertas como números. Se tiene que revisar en el contexto del problema cuáles son realmente numéricas.

Diccionarios

```
Dicc19 <- read_csv(here("data", "Diccionario_molec_2019.csv"))
```

```

Rows: 108 Columns: 6
— Column specification

```

```
Delimiter: ","
```

```
chr (5): NOMBRE_CAMPO, TIPO, MNEMONICO, CATÁLOGO, RANGO_CLAVES
```

```
dbl (1): LONGITUD
```

i Use ``spec()`` to retrieve the full column specification for this data.
i Specify the column types or set ``show_col_types = FALSE`` to quiet this message.

El diccionario indica qué significan las variables en el contexto del problema, pero no a qué equivalen sus categorías. Eso se debe leer en el archivo de descriptor de las preguntas y en el cuestionario.

Resumen de las variables

```
summary(M19)
```

cd	periodo	folio	entidad
Length:1980	Length:1980	Length:1980	Length:1980
Class :character	Class :character	Class :character	Class :character
Mode :character	Mode :character	Mode :character	Mode :character

control	viv_sel	num_hog	hog_mud
Min. :40001	Min. :1.000	Min. :1	Min. :0.00000
1st Qu.:40138	1st Qu.:2.000	1st Qu.:1	1st Qu.:0.00000
Median :40222	Median :3.000	Median :1	Median :0.00000
Mean :40291	Mean :2.502	Mean :1	Mean :0.04697
3rd Qu.:40390	3rd Qu.:3.000	3rd Qu.:1	3rd Qu.:0.00000
Max. :41420	Max. :4.000	Max. :1	Max. :2.00000

n_ren_ele	p1	p2	p3_1
Length:1980	Min. :1.00	Min. :0.000	Min. :0.00
Class :character	1st Qu.:1.00	1st Qu.:1.000	1st Qu.:1.00
Mode :character	Median :1.00	Median :1.000	Median :2.00
	Mean :1.03	Mean :1.351	Mean :1.53
	3rd Qu.:1.00	3rd Qu.:2.000	3rd Qu.:2.00
	Max. :2.00	Max. :2.000	Max. :2.00

p3_2	p3_3	p3_4	p3_5	p4
Min. :0.000	Min. :0.0	Min. :0.000	Min. :0.000	Min. : 0.000
1st Qu.:1.000	1st Qu.:1.0	1st Qu.:2.000	1st Qu.:1.000	1st Qu.: 0.000
Median :2.000	Median :2.0	Median :2.000	Median :2.000	Median : 0.000
Mean :1.612	Mean :1.6	Mean :1.897	Mean :1.582	Mean : 1.335
3rd Qu.:2.000	3rd Qu.:2.0	3rd Qu.:2.000	3rd Qu.:2.000	3rd Qu.: 2.000
Max. :2.000	Max. :2.0	Max. :2.000	Max. :2.000	Max. :50.000

p5	p5_6esp	p6_1	p6_2
Min. :0.00	Length:1980	Min. :0.0000	Min. :0.0000
1st Qu.:0.00	Class :character	1st Qu.:0.0000	1st Qu.:0.0000

Median :0.00	Mode :character	Median :0.0000	Median :0.0000
Mean :1.37		Mean :0.7848	Mean :0.6889
3rd Qu.:3.00		3rd Qu.:2.0000	3rd Qu.:2.0000
Max. :6.00		Max. :2.0000	Max. :2.0000

p6_3	p6_4	p6_5	p6_6
Min. :0.0000	Min. :0.0000	Min. :0.0000	Min. :0.0000
1st Qu.:0.0000	1st Qu.:0.0000	1st Qu.:0.0000	1st Qu.:0.0000
Median :0.0000	Median :0.0000	Median :0.0000	Median :0.0000
Mean :0.6985	Mean :0.6465	Mean :0.7131	Mean :0.8177
3rd Qu.:2.0000	3rd Qu.:1.0000	3rd Qu.:2.0000	3rd Qu.:2.0000
Max. :2.0000	Max. :2.0000	Max. :2.0000	Max. :2.0000

p6_6esp	p7	p7_3	p8_1
Mode:logical	Min. :0.0000	Min. : 0.0	Min. :0.0000
NA's:1980	1st Qu.:0.0000	1st Qu.: 0.0	1st Qu.:0.0000
	Median :0.0000	Median : 0.0	Median :0.0000
	Mean :0.9364	Mean : 195.5	Mean :0.7545
	3rd Qu.:2.0000	3rd Qu.: 0.0	3rd Qu.:2.0000
	Max. :3.0000	Max. :99999.0	Max. :2.0000

p8_2	p9	p9_5esp	p10
Min. :0.0000	Min. :0.0000	Length:1980	Min. : 0.000
1st Qu.:0.0000	1st Qu.:0.0000	Class :character	1st Qu.: 0.000
Median :0.0000	Median :0.0000	Mode :character	Median : 0.000
Mean :0.4561	Mean :0.4848		Mean : 1.363
3rd Qu.:1.0000	3rd Qu.:1.0000		3rd Qu.: 2.000
Max. :2.0000	Max. :5.0000		Max. :99.000

p11	p11_6esp	p12_1	p12_2
Min. :0.000	Mode:logical	Min. :0.0000	Min. :0.0000
1st Qu.:0.000	NA's:1980	1st Qu.:0.0000	1st Qu.:0.0000
Median :0.000		Median :0.0000	Median :0.0000
Mean :1.156		Mean :0.6197	Mean :0.5955
3rd Qu.:3.000		3rd Qu.:2.0000	3rd Qu.:2.0000
Max. :6.000		Max. :2.0000	Max. :2.0000

p12_3	p12_4	p12_5	p12_6
Min. :0.0000	Min. :0.0000	Min. :0.0000	Min. :0.0000
1st Qu.:0.0000	1st Qu.:0.0000	1st Qu.:0.0000	1st Qu.:0.0000
Median :0.0000	Median :0.0000	Median :0.0000	Median :0.0000
Mean :0.5949	Mean :0.5742	Mean :0.6172	Mean :0.5818
3rd Qu.:2.0000	3rd Qu.:1.0000	3rd Qu.:2.0000	3rd Qu.:2.0000
Max. :2.0000	Max. :2.0000	Max. :2.0000	Max. :2.0000

p12_7	p12_8	p12_9	p12_9esp
Min. :0.0000	Min. :0.000	Min. :0.000	Length:1980
1st Qu.:0.0000	1st Qu.:0.000	1st Qu.:0.000	Class :character
Median :0.0000	Median :0.000	Median :0.000	Mode :character

Mean :0.5773	Mean :0.554	Mean :0.651
3rd Qu.:1.2500	3rd Qu.:1.000	3rd Qu.:2.000
Max. :2.0000	Max. :2.000	Max. :2.000

p13	p13_3	p14_1	p14_2
Min. :0.000	Min. : 0.00	Min. :0.0000	Min. :0.0000
1st Qu.:0.000	1st Qu.: 0.00	1st Qu.:0.0000	1st Qu.:0.0000
Median :0.000	Median : 0.00	Median :0.0000	Median :0.0000
Mean :0.752	Mean : 18.22	Mean :0.6197	Mean :0.3551
3rd Qu.:2.000	3rd Qu.: 0.00	3rd Qu.:2.0000	3rd Qu.:1.0000
Max. :3.000	Max. :1000.00	Max. :2.0000	Max. :2.0000

p15	p15_5esp	p16	p17
Min. :0.0000	Length:1980	Min. : 0.000	Min. :0.00
1st Qu.:0.0000	Class :character	1st Qu.: 0.000	1st Qu.:0.00
Median :0.0000	Mode :character	Median : 0.000	Median :0.00
Mean :0.4096		Mean : 1.092	Mean :1.11
3rd Qu.:1.0000		3rd Qu.: 1.000	3rd Qu.:3.00
Max. :5.0000		Max. :45.000	Max. :6.00

p17_6esp	p18_1	p18_2	p18_3
Length:1980	Min. :0.0000	Min. :0.0000	Min. :0.0000
Class :character	1st Qu.:0.0000	1st Qu.:0.0000	1st Qu.:0.0000
Mode :character	Median :0.0000	Median :0.0000	Median :0.0000
	Mean :0.4268	Mean :0.4692	Mean :0.4611
	3rd Qu.:1.0000	3rd Qu.:1.0000	3rd Qu.:1.0000
	Max. :2.0000	Max. :2.0000	Max. :2.0000

p18_4	p18_5	p19	p19_3
Min. :0.0000	Min. :0.000	Min. :0.0000	Min. : 0.0
1st Qu.:0.0000	1st Qu.:0.000	1st Qu.:0.0000	1st Qu.: 0.0
Median :0.0000	Median :0.000	Median :0.0000	Median : 0.0
Mean :0.4975	Mean :0.403	Mean :0.8177	Mean : 512.7
3rd Qu.:1.0000	3rd Qu.:1.000	3rd Qu.:2.0000	3rd Qu.: 0.0
Max. :2.0000	Max. :2.000	Max. :3.0000	Max. :999999.0

p20_1	p20_2	p21	p21_5esp
Min. :0.0000	Min. :0.0000	Min. :0.0000	Length:1980
1st Qu.:0.0000	1st Qu.:0.0000	1st Qu.:0.0000	Class :character
Median :0.0000	Median :0.0000	Median :0.0000	Mode :character
Mean :0.6475	Mean :0.3672	Mean :0.4879	
3rd Qu.:2.0000	3rd Qu.:1.0000	3rd Qu.:1.0000	
Max. :2.0000	Max. :2.0000	Max. :5.0000	

p22	p23_1	p23_2	p24
Min. :0.0000	Min. :0.00000	Min. :0.00000	Min. :0.0000
1st Qu.:0.0000	1st Qu.:0.00000	1st Qu.:0.00000	1st Qu.:0.0000
Median :0.0000	Median :0.00000	Median :0.00000	Median :0.0000
Mean :0.1374	Mean :0.07121	Mean :0.05253	Mean :0.7631

3rd Qu.:0.0000	3rd Qu.:0.00000	3rd Qu.:0.00000	3rd Qu.:1.0000
Max. :5.0000	Max. :2.00000	Max. :2.00000	Max. :5.0000

p25	p25_6esp	p26	p27
Min. :0.000	Length:1980	Min. : 0.00	Min. :0.000
1st Qu.:0.000	Class :character	1st Qu.: 0.00	1st Qu.:0.000
Median :0.000	Mode :character	Median : 20.00	Median :2.000
Mean :1.066		Mean : 27.54	Mean :1.322
3rd Qu.:3.000		3rd Qu.: 40.00	3rd Qu.:2.000
Max. :6.000		Max. :360.00	Max. :2.000

p28	p28_7esp	p29	p30
Min. :0.0000	Length:1980	Min. :0.000	Min. :0.000
1st Qu.:0.0000	Class :character	1st Qu.:0.000	1st Qu.:0.000
Median :0.0000	Mode :character	Median :2.000	Median :3.000
Mean :0.5808		Mean :1.748	Mean :2.111
3rd Qu.:0.0000		3rd Qu.:3.000	3rd Qu.:3.000
Max. :7.0000		Max. :4.000	Max. :4.000

p31	p32	p32_6esp	p33_1
Min. :0.000	Min. :0.0000	Length:1980	Min. :0.000
1st Qu.:0.000	1st Qu.:0.0000	Class :character	1st Qu.:1.000
Median :1.000	Median :0.0000	Mode :character	Median :2.000
Mean :1.134	Mean :0.7106		Mean :1.712
3rd Qu.:2.000	3rd Qu.:1.0000		3rd Qu.:2.000
Max. :2.000	Max. :6.0000		Max. :2.000

p33_2	p33_3	p33_4	p34_1	p34_2
Min. :0.000	Min. :0.000	Min. :0.000	Min. :0.00	Min. :0.00
1st Qu.:2.000	1st Qu.:2.000	1st Qu.:2.000	1st Qu.:1.00	1st Qu.:1.00
Median :2.000	Median :2.000	Median :2.000	Median :2.00	Median :1.00
Mean :1.801	Mean :1.856	Mean :1.767	Mean :1.71	Mean :1.47
3rd Qu.:2.000	3rd Qu.:2.000	3rd Qu.:2.000	3rd Qu.:2.00	3rd Qu.:2.00
Max. :2.000	Max. :2.000	Max. :2.000	Max. :3.00	Max. :3.00

p34_3	p34_3_1	p34_4	p34_4_1
Min. :0.000	Min. :0.0000	Min. :0.000	Min. :0.0000
1st Qu.:1.000	1st Qu.:0.0000	1st Qu.:1.000	1st Qu.:0.0000
Median :2.000	Median :0.0000	Median :1.000	Median :1.0000
Mean :1.639	Mean :0.5323	Mean :1.416	Mean :0.9621
3rd Qu.:2.000	3rd Qu.:1.0000	3rd Qu.:2.000	3rd Qu.:1.0000
Max. :3.000	Max. :3.0000	Max. :3.000	Max. :6.0000

p35	p36_1	p36_2	p36_3
Min. :0.0000	Min. :0.000	Min. :0.000	Min. :0.000
1st Qu.:1.0000	1st Qu.:1.000	1st Qu.:1.000	1st Qu.:1.000
Median :1.0000	Median :1.000	Median :1.000	Median :1.000
Mean :0.9818	Mean :1.424	Mean :1.207	Mean :1.334
3rd Qu.:1.0000	3rd Qu.:2.000	3rd Qu.:1.000	3rd Qu.:2.000

```

Max.      :2.0000   Max.      :3.000   Max.      :3.000   Max.      :3.000

      p36_4          factor          h_lec          mat_lec
Min.      :0.000   Min.      : 2852   Min.      :1.000   Min.      :1.000
1st Qu.:1.000   1st Qu.: 12880   1st Qu.:1.000   1st Qu.:2.000
Median :1.000   Median : 16764   Median :1.000   Median :3.000
Mean    :1.422   Mean    : 19297   Mean    :2.044   Mean    :2.756
3rd Qu.:2.000   3rd Qu.: 24724   3rd Qu.:3.000   3rd Qu.:4.000
Max.    :3.000   Max.    :103721   Max.    :4.000   Max.    :4.000
NA's    :60      NA's    :60

      perslec      l_formato      r_formato      p_formato      persleclrp
Min.      :1.000   Min.      :1.000   Min.      :1.000   Min.      :1.000   Min.      :1.0
1st Qu.:1.000   1st Qu.:2.000   1st Qu.:2.000   1st Qu.:2.000   1st Qu.:1.0
Median :1.000   Median :2.000   Median :2.000   Median :2.000   Median :1.0
Mean    :1.258   Mean    :1.926   Mean    :1.937   Mean    :1.929   Mean    :1.3
3rd Qu.:2.000   3rd Qu.:2.000   3rd Qu.:2.000   3rd Qu.:2.000   3rd Qu.:2.0
Max.    :2.000   Max.    :3.000   Max.    :3.000   Max.    :3.000   Max.    :2.0
NA's    :60      NA's    :1170   NA's    :1332   NA's    :1308   NA's    :60

      edad      sexo      anio      nivel
Min.      :18.00   Min.      :1.000   Min.      :1.000   Min.      : 0.000
1st Qu.:31.00   1st Qu.:1.000   1st Qu.:3.000   1st Qu.: 3.000
Median :44.00   Median :2.000   Median :3.000   Median : 4.000
Mean    :45.07   Mean    :1.559   Mean    :3.463   Mean    : 4.368
3rd Qu.:57.00   3rd Qu.:2.000   3rd Qu.:4.000   3rd Qu.: 7.000
Max.    :95.00   Max.    :2.000   Max.    :6.000   Max.    :99.000
NA's    :59

      cond_activ
Length:1980
Class :character
Mode  :character

```

Aunque en la mayor parte de las variables, las medidas calculadas (media, mediana, mínimo, máximo), en muchas no tienen sentido porque son categóricas. El summary permite ver si la variable tiene datos faltantes (NA's), por ejemplo, la variable anio tiene valores faltantes.

Selección de las variables

Por ejemplo, si me interesa analizar las variables 3 a 34 (se revisa en la base de datos el número de columna que tiene la variable de interés):

```

M19selec <- M19[3:34]
M20selec <- M20[3:34]

```

Datos perdidos o faltantes

Visualizo si las variables de interés tienen datos perdidos:

```
summary(M20selec)
```

control	viv_sel	num_hog	hog_mud	n_ren_el
Min. :40025	Min. :1.000	Min. :1	Min. :0.00000	Min. : 1.000
1st Qu.:40153	1st Qu.:2.000	1st Qu.:1	1st Qu.:0.00000	1st Qu.: 1.000
Median :40248	Median :3.000	Median :1	Median :0.00000	Median : 1.000
Mean :40307	Mean :2.517	Mean :1	Mean :0.04726	Mean : 1.734
3rd Qu.:40398	3rd Qu.:4.000	3rd Qu.:1	3rd Qu.:0.00000	3rd Qu.: 2.000
Max. :41420	Max. :4.000	Max. :1	Max. :2.00000	Max. :10.000

cd	periodo	sexo	edad	anio
Min. : 1.00	Min. :220	Min. :1.000	Min. :18.00	Min. :1.00
1st Qu.: 2.00	1st Qu.:220	1st Qu.:1.000	1st Qu.:31.00	1st Qu.:3.00
Median : 9.00	Median :220	Median :2.000	Median :44.00	Median :3.00
Mean :14.01	Mean :220	Mean :1.552	Mean :45.49	Mean :3.47
3rd Qu.:25.00	3rd Qu.:220	3rd Qu.:2.000	3rd Qu.:58.00	3rd Qu.:4.00
Max. :43.00	Max. :220	Max. :2.000	Max. :94.00	Max. :6.00

NA's :65

nivel	cond_act	p1	p2
Min. : 0.000	Min. : 1.000	Min. :1.000	Min. :0.000
1st Qu.: 3.000	1st Qu.: 1.000	1st Qu.:1.000	1st Qu.:1.000
Median : 4.000	Median : 1.000	Median :1.000	Median :1.000
Mean : 4.418	Mean : 3.289	Mean :1.026	Mean :1.367
3rd Qu.: 7.000	3rd Qu.: 7.000	3rd Qu.:1.000	3rd Qu.:2.000
Max. :99.000	Max. :99.000	Max. :2.000	Max. :2.000

p3_1	p3_2	p3_3	p3_4
Min. :0.000	Min. :0.000	Min. :0.000	Min. :0.000
1st Qu.:1.000	1st Qu.:1.000	1st Qu.:1.000	1st Qu.:2.000
Median :2.000	Median :2.000	Median :2.000	Median :2.000
Mean :1.545	Mean :1.638	Mean :1.655	Mean :1.904
3rd Qu.:2.000	3rd Qu.:2.000	3rd Qu.:2.000	3rd Qu.:2.000
Max. :2.000	Max. :2.000	Max. :2.000	Max. :2.000

p3_5	p4	p5	p5_6esp
Min. :0.000	Min. : 0.000	Min. :0.00	Length:2010
1st Qu.:1.000	1st Qu.: 0.000	1st Qu.:0.00	Class :character
Median :2.000	Median : 0.000	Median :0.00	Mode :character
Mean :1.588	Mean : 1.428	Mean :1.33	
3rd Qu.:2.000	3rd Qu.: 2.000	3rd Qu.:3.00	
Max. :2.000	Max. :60.000	Max. :6.00	

p6_1	p6_2	p6_3	p6_4
Min. :0.0000	Min. :0.0000	Min. :0.0000	Min. :0.0000
1st Qu.:0.0000	1st Qu.:0.0000	1st Qu.:0.0000	1st Qu.:0.0000

Median :0.0000	Median :0.0000	Median :0.0000	Median :0.0000
Mean :0.7781	Mean :0.6801	Mean :0.6861	Mean :0.6363
3rd Qu.:2.0000	3rd Qu.:2.0000	3rd Qu.:2.0000	3rd Qu.:1.0000
Max. :2.0000	Max. :2.0000	Max. :2.0000	Max. :2.0000

p6_5	p6_6	p6_6esp	p7
Min. :0.000	Min. :0.000	Length:2010	Min. :0.0000
1st Qu.:0.000	1st Qu.:0.000	Class :character	1st Qu.:0.0000
Median :0.000	Median :0.000	Mode :character	Median :0.0000
Mean :0.696	Mean :0.805		Mean :0.9224
3rd Qu.:2.000	3rd Qu.:2.000		3rd Qu.:2.0000
Max. :2.000	Max. :2.000		Max. :3.0000

p7_3	p8_1
Min. : 0	Min. :0.0000
1st Qu.: 0	1st Qu.:0.0000
Median : 0	Median :0.0000
Mean : 1631	Mean :0.7473
3rd Qu.: 0	3rd Qu.:2.0000
Max. :999999	Max. :2.0000

La variable anio tiene 39 datos faltantes. Anio está en la columna 10 en la base de datos de las variables de interés.

Para calcular el porcentaje de datos faltantes que tiene y que se los vamos a quitar:

```
sum(is.na(M20selec[, 10])) / length(M20selec[, 10])

[1] 65
```

Para quitar los NA's

```
M20selec <- na.omit(M20selec)
summary(M20selec)
```

control	viv_sel	num_hog	hog_mud	n_ren_el
Min. : NA	Min. : NA	Min. : NA	Min. : NA	Min. : NA
1st Qu.: NA	1st Qu.: NA	1st Qu.: NA	1st Qu.: NA	1st Qu.: NA
Median : NA	Median : NA	Median : NA	Median : NA	Median : NA
Mean :NaN	Mean :NaN	Mean :NaN	Mean :NaN	Mean :NaN
3rd Qu.: NA	3rd Qu.: NA	3rd Qu.: NA	3rd Qu.: NA	3rd Qu.: NA
Max. : NA	Max. : NA	Max. : NA	Max. : NA	Max. : NA
cd	periodo	sexo	edad	anio
Min. : NA	Min. : NA	Min. : NA	Min. : NA	Min. : NA
1st Qu.: NA	1st Qu.: NA	1st Qu.: NA	1st Qu.: NA	1st Qu.: NA
Median : NA	Median : NA	Median : NA	Median : NA	Median : NA
Mean :NaN	Mean :NaN	Mean :NaN	Mean :NaN	Mean :NaN
3rd Qu.: NA	3rd Qu.: NA	3rd Qu.: NA	3rd Qu.: NA	3rd Qu.: NA
Max. : NA	Max. : NA	Max. : NA	Max. : NA	Max. : NA
nivel	cond_act	p1	p2	p3_1

Min. : NA	Min. : NA	Min. : NA	Min. : NA	Min. : NA
1st Qu.: NA	1st Qu.: NA	1st Qu.: NA	1st Qu.: NA	1st Qu.: NA
Median : NA	Median : NA	Median : NA	Median : NA	Median : NA
Mean :NaN	Mean :NaN	Mean :NaN	Mean :NaN	Mean :NaN
3rd Qu.: NA	3rd Qu.: NA	3rd Qu.: NA	3rd Qu.: NA	3rd Qu.: NA
Max. : NA	Max. : NA	Max. : NA	Max. : NA	Max. : NA
p3_2	p3_3	p3_4	p3_5	p4
Min. : NA	Min. : NA	Min. : NA	Min. : NA	Min. : NA
1st Qu.: NA	1st Qu.: NA	1st Qu.: NA	1st Qu.: NA	1st Qu.: NA
Median : NA	Median : NA	Median : NA	Median : NA	Median : NA
Mean :NaN	Mean :NaN	Mean :NaN	Mean :NaN	Mean :NaN
3rd Qu.: NA	3rd Qu.: NA	3rd Qu.: NA	3rd Qu.: NA	3rd Qu.: NA
Max. : NA	Max. : NA	Max. : NA	Max. : NA	Max. : NA
p5	p5_6esp	p6_1	p6_2	p6_3
Min. : NA	Length:0	Min. : NA	Min. : NA	Min. : NA
1st Qu.: NA	Class :character	1st Qu.: NA	1st Qu.: NA	1st Qu.: NA
Median : NA	Mode :character	Median : NA	Median : NA	Median : NA
Mean :NaN		Mean :NaN	Mean :NaN	Mean :NaN
3rd Qu.: NA		3rd Qu.: NA	3rd Qu.: NA	3rd Qu.: NA
Max. : NA		Max. : NA	Max. : NA	Max. : NA
p6_4	p6_5	p6_6	p6_6esp	p7
Min. : NA	Min. : NA	Min. : NA	Length:0	Min. : NA
1st Qu.: NA	1st Qu.: NA	1st Qu.: NA	Class :character	1st Qu.: NA
Median : NA	Median : NA	Median : NA	Mode :character	Median : NA
Mean :NaN	Mean :NaN	Mean :NaN		Mean :NaN
3rd Qu.: NA	3rd Qu.: NA	3rd Qu.: NA		3rd Qu.: NA
Max. : NA	Max. : NA	Max. : NA		Max. : NA
p7_3	p8_1			
Min. : NA	Min. : NA			
1st Qu.: NA	1st Qu.: NA			
Median : NA	Median : NA			
Mean :NaN	Mean :NaN			
3rd Qu.: NA	3rd Qu.: NA			
Max. : NA	Max. : NA			

Se verifica que ya no hay datos perdidos en mi data set de interés.

prueba